

String Theory and M-Theory

Lecture 6: Scattering, Duality, and the Veneziano Amplitude

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Chapter 1

Scattering, Duality, and the Veneziano Amplitude

Overview. Scattering reduces a collision to invariant data that can be compared with experiment. For four identical particles those data may be organized into the Mandelstam variables s, t, u , only two of which are independent. Ordinary particle exchange then suggests separate direct and crossed channels. The Veneziano amplitude overturned that expectation: one function contains both channel descriptions. The second half of the lecture builds that function from two open strings that join, evolve as one oscillating chain, and split again.

1.1 What a Theory Must Predict

Fermionic strings solve two defects of the purely bosonic model: they admit fermionic states and, after the appropriate projection, remove the tachyon. Their critical spacetime dimension is ten rather than twenty-six. Extra dimensions therefore remain, but their compactification is postponed. The immediate subject is scattering.¹

Scattering is not necessarily the most interesting thing particles do. It is simply the operation that an experiment can repeat cleanly. Send specified particles toward one another, record the particles emerging from the collision, and infer the dynamics hidden inside the interaction region. A theory becomes quantitative when it supplies a complex amplitude

$$\mathcal{A}(\text{incoming data} \rightarrow \text{outgoing data}), \quad (1.1)$$

whose squared magnitude, together with the usual phase-space factors, gives the probability for that outcome.

¹Editorial clarification: The tachyon-free statement refers to the consistent superstring spectrum after the required projection; merely adding world-sheet fermions is not by itself enough.

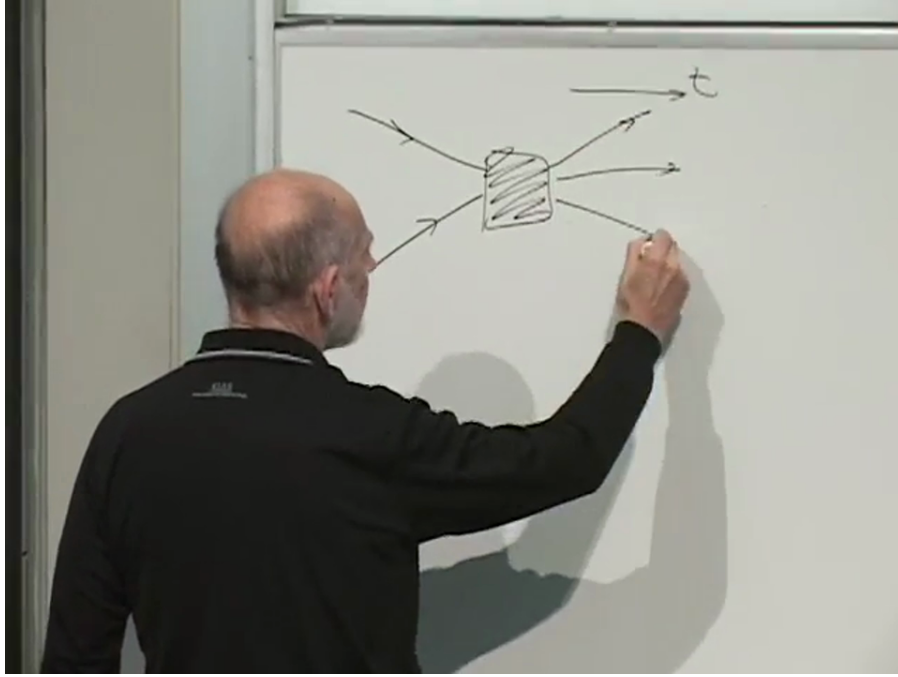


Figure 1.1: Incoming particles enter a black-box interaction region and outgoing particles emerge; time is drawn horizontally.

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Particles also carry spin, charge, and species labels. To expose the kinematics first, let us temporarily suppress everything except momentum.

1.2 Four-Momentum and the All-Incoming Convention

Write the four-momentum of one particle as

$$k^\mu = (k^0, \mathbf{k}) = (E, \mathbf{p}). \quad (1.2)$$

The lecture uses the mostly-plus metric, so

$$k^2 \equiv \mathbf{p}^2 - E^2. \quad (1.3)$$

The relativistic dispersion relation is therefore

$$E^2 - \mathbf{p}^2 = m^2, \quad k^2 = -m^2. \quad (1.4)$$

Changing the spatial momentum changes the energy, but it does not move the particle off its mass shell.

$$\begin{aligned}
 E &= p + m \\
 E^2 - p^2 &= m^2 \\
 p^2 - E^2 &= -m^2 \\
 p_\mu p^\mu &= k^2
 \end{aligned}$$

Figure 1.2: The mass-shell condition in the metric convention used throughout the lecture.

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Consider two particles entering with momenta k_1, k_2 and two leaving with physical momenta q_3, q_4 . Conservation of energy and momentum is one four-vector equation,

$$k_1 + k_2 = q_3 + q_4. \quad (1.5)$$

It is convenient to assign an incoming momentum to every external line:

$$k_3 \equiv -q_3, \quad k_4 \equiv -q_4. \quad (1.6)$$

Then

$$k_1 + k_2 + k_3 + k_4 = 0, \quad (1.7)$$

and every leg obeys $k_i^2 = -m_i^2$. The outgoing particles have negative energy only in this bookkeeping convention; their physical energies remain positive.

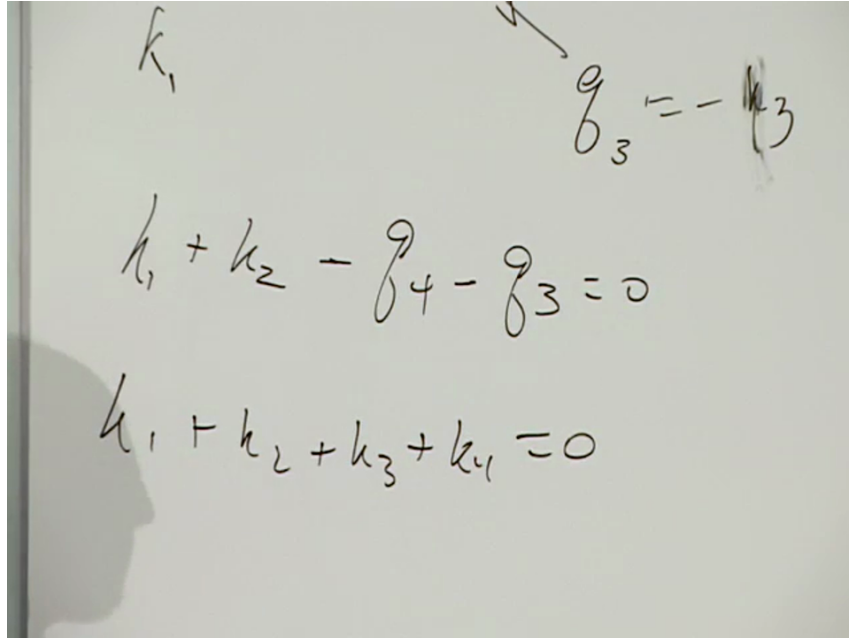


Figure 1.3: Physical outgoing momenta are reversed so that all four external momenta may be treated symmetrically as incoming.

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At first the amplitude appears to be a function of sixteen components, $\mathcal{A}(k_1, k_2, k_3, k_4)$. Most of them are redundant. Go to the center-of-mass frame, rotate the incoming momenta onto a chosen axis, and assume for simplicity that all four particles have the same mass. The incoming state is then fixed by one number, the center-of-mass energy. The outgoing spatial momenta must be equal and opposite and have the same magnitude as the incoming pair. Only their deflection angle θ remains. The process therefore has two independent kinematic variables:

$$E_{\text{cm}}, \quad \theta. \quad (1.8)$$

1.3 Mandelstam Variables

The two physical quantities in (1.8) should be expressible without choosing a frame. With the all-incoming convention define

$$-s = (k_1 + k_2)^2, \quad -t = (k_1 + k_3)^2, \quad -u = (k_1 + k_4)^2. \quad (1.9)$$

In the center-of-mass frame the spatial parts of $k_1 + k_2$ cancel, hence

$$s = E_{\text{cm}}^2. \quad (1.10)$$

For an outgoing physical momentum $q_3 = -k_3$,

$$k_1 + k_3 = k_1 - q_3 \quad (1.11)$$

is the momentum transferred from leg 1 to leg 3. For equal external masses, with incoming spatial momentum magnitude p ,

$$-t = 2p^2(1 - \cos \theta), \quad -u = 2p^2(1 + \cos \theta). \quad (1.12)$$

Thus t and u carry the angular information measured by detectors.

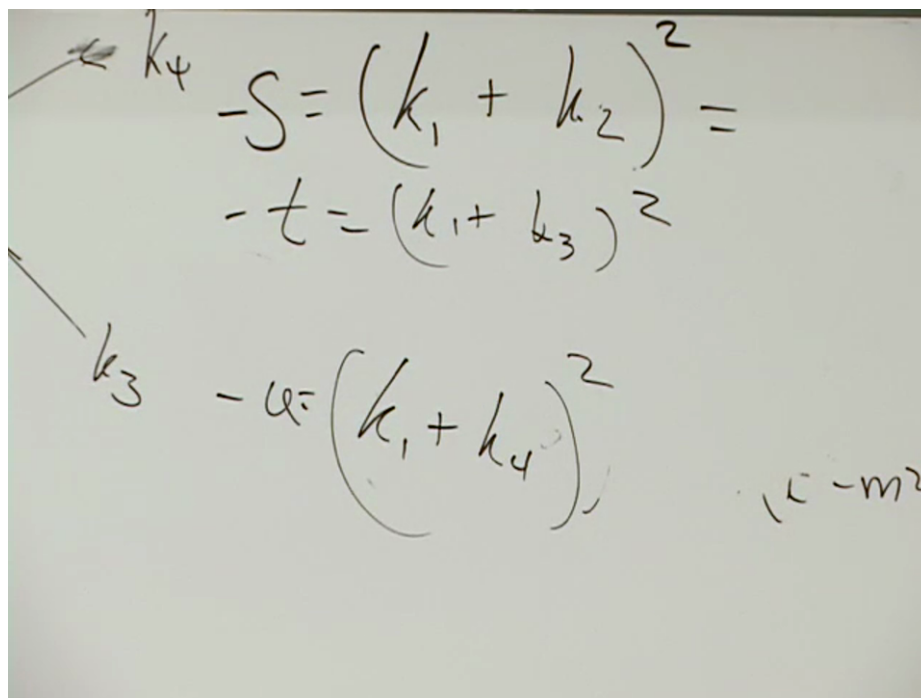


Figure 1.4: The symmetric definitions of s, t, u in the all-incoming convention.

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Three symbols do not mean three independent data. Squaring (1.7) and using $k_i^2 = -m^2$ gives

$$s + t + u = 4m^2 \quad (1.13)$$

for equal masses. Once s and t are known, u is fixed.

Question from the audience. Could the vacuum slow a freely moving particle, rather like friction? ^a

Response. Not in a translation-invariant vacuum. There is nowhere for the missing momentum to go, so momentum conservation forbids such friction. One can imagine a medium or a different vacuum that selects a preferred rest frame, but that is not the vacuum being used here.

^aLecture timestamp: 00:27:17.

Question from the audience. Why must the two outgoing particles have equal energies? ^a

Response. In the center-of-mass frame their spatial momenta must be equal and opposite. For the equal-mass process under discussion, equal momentum magnitudes give equal energies through $E^2 = p^2 + m^2$. If the outgoing masses differed, that last conclusion would have to be modified.

^aLecture timestamp: 00:27:54.

1.4 Direct and Crossed Channels

A simple point-particle diagram lets particles 1 and 2 form an intermediate particle of mass M , which then decays into 3 and 4. Up to numerator factors and the pole prescription, its characteristic dependence is

$$\mathcal{A}_s \sim \frac{g^2}{s - M^2}. \quad (1.14)$$

This is the direct, or s -channel, process. Turning the connectivity sideways gives exchange between the two external lines,

$$\mathcal{A}_t \sim \frac{g^2}{t - M^2}, \quad (1.15)$$

the crossed, or t -channel, process. A third connectivity gives

$$\mathcal{A}_u \sim \frac{g^2}{u - M^2}. \quad (1.16)$$

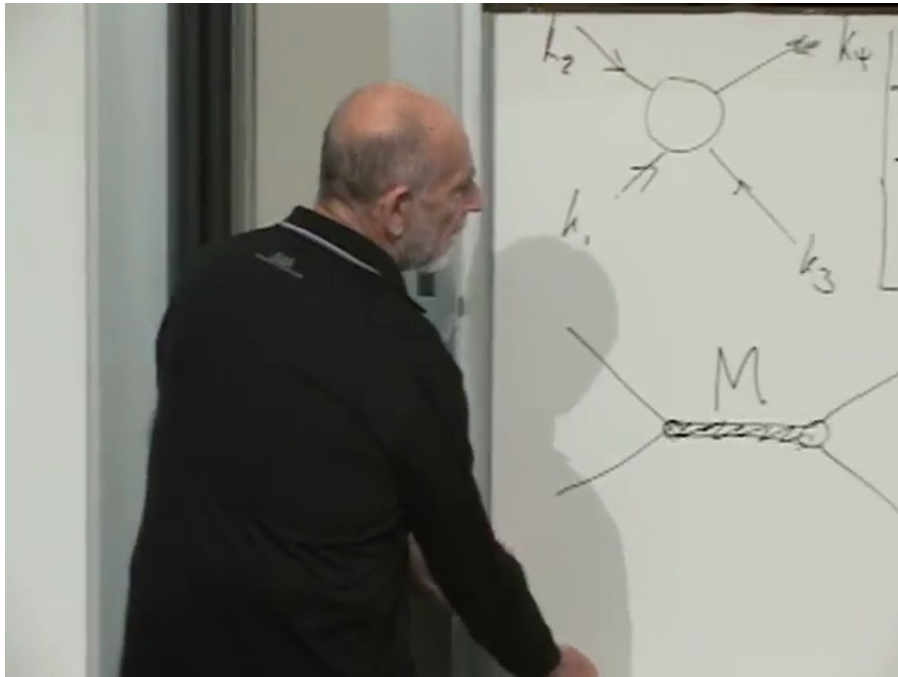


Figure 1.5: Direct production through an intermediate state and the corresponding pole $g^2/(s - M^2)$; the crossed connectivity is being assembled on the same board.

Lecture frame: 00:30:50.

The distinction is physical. At this crude scalar level, (1.14) depends on collision energy but not on angle. The compound state has forgotten the incident direction before it decays, so its decay is isotropic. The exchange amplitude depends on momentum transfer and therefore on angle; a light exchanged particle strongly favors small deflection. Spin would add further angular dependence, but the pole structure is already visible.

Question from the audience. Does the missing u -channel produce a third term? ^a

Response. Yes, whenever the particle identities and couplings permit that connectivity. One then includes a term with a pole at $u = M^2$. It is not a new independent kinematic variable, because $s + t + u = 4m^2$, but it is a distinct channel diagram.

^aLecture timestamp: 00:34:47.

1.5 One Function, Two Channel Descriptions

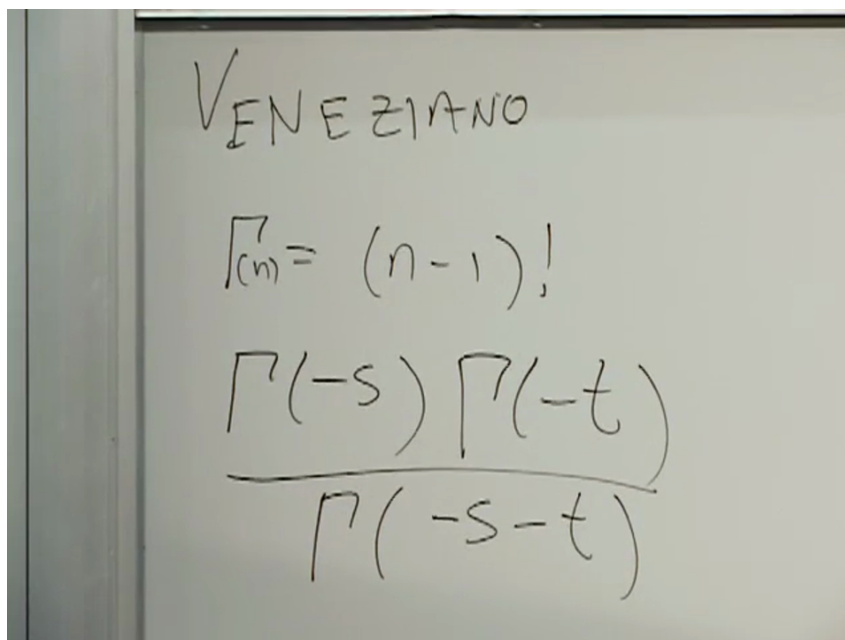
A single pole is a poor model of meson scattering. Two mesons can form many resonances, with a tower of masses and angular momenta. The first response in the 1960s was to add more s -channel poles, more t -channel poles, and the allowed u -channel poles. Then a much more economical function was found. In the lecture's units, where the linear Regge trajectory and its intercept have been absorbed into s and t , it is

$$\mathcal{A}_V(s, t) \propto \frac{\Gamma(-s)\Gamma(-t)}{\Gamma(-s-t)}. \quad (1.17)$$

The gamma function continues the factorial:

$$\Gamma(n) = (n-1)! \quad (1.18)$$

at positive integers.



A photograph of a whiteboard with handwritten text. At the top, the word 'VENEZIANO' is written in large, capital letters. Below it, the equation $\Gamma(n) = (n-1)!$ is written. At the bottom, the Veneziano amplitude is written as a fraction: $\frac{\Gamma(-s)\Gamma(-t)}{\Gamma(-s-t)}$.

Figure 1.6: The Veneziano amplitude written as a symmetric ratio of gamma functions.

Lecture frame: 00:42:25.

The poles of $\Gamma(-s)$ reproduce an infinite tower of direct-channel resonances. Yet the same expression is symmetric under $s \leftrightarrow t$, so its $\Gamma(-t)$ poles give a crossed-channel expansion. These are not two separately added contributions. One analytic function admits either description. This is the old idea of duality between direct-channel resonances and crossed-channel exchanges.

Question from the audience. How can one amplitude contain both channel sums without adding them? ^a

Response. That is precisely the surprise. Expanding the function near its s -channel poles organizes it as direct resonances; expanding it near its t -channel poles organizes the same function as exchanges. A microscopic model is needed to explain why those two readings are the same, and the model is an interacting string.

^aLecture timestamp: 00:43:32.

1.6 A Strip Swept Out by an Open String

Parameterize an open string by

$$0 \leq \sigma \leq \pi. \quad (1.19)$$

The label σ is not a spacetime position. The embedding $X^\mu(\sigma, \tau)$ tells us where that labeled point lies in spacetime, while τ measures evolution. A point particle sweeps out a worldline; the moving string sweeps out a two-dimensional worldsheet. In the free theory each embedding coordinate obeys

$$\frac{\partial^2 X^\mu}{\partial \tau^2} - \frac{\partial^2 X^\mu}{\partial \sigma^2} = 0. \quad (1.20)$$

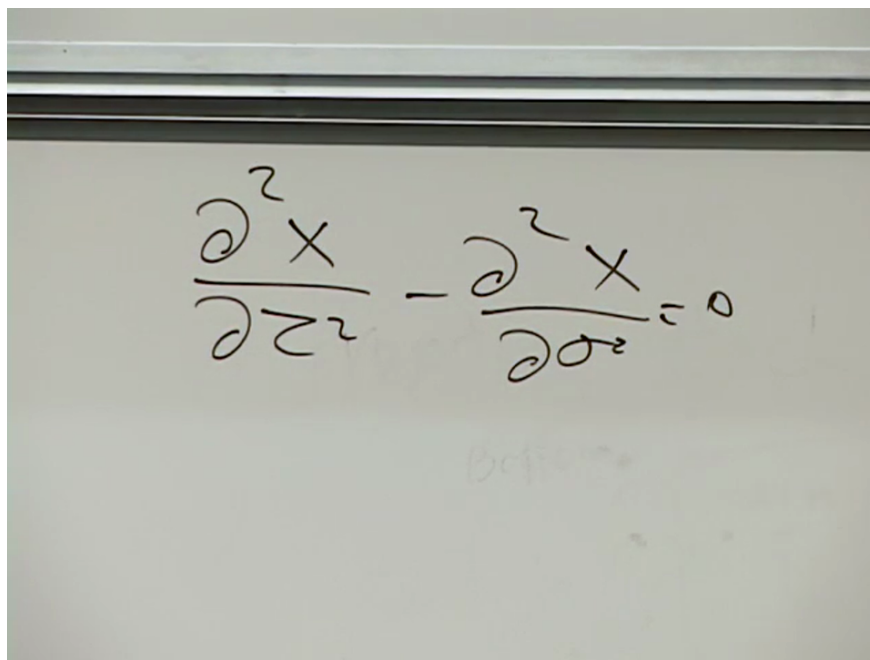


Figure 1.7: The wave equation governing each free string coordinate on the worldsheet strip.

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For the scattering construction it is useful not to take the continuum limit too soon. Replace each string by n mass points joined by springs. The worldsheet becomes a dense collection of nearby worldlines, and its normal modes are ordinary harmonic oscillators. Two incoming chains can touch at their endpoints. At that event a new spring joins them, they evolve for a time τ as one longer chain, and later a spring breaks so that two outgoing chains emerge.

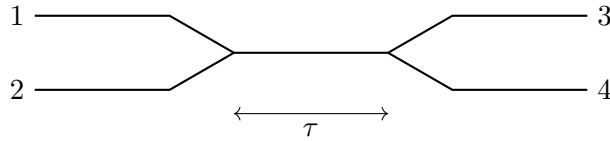
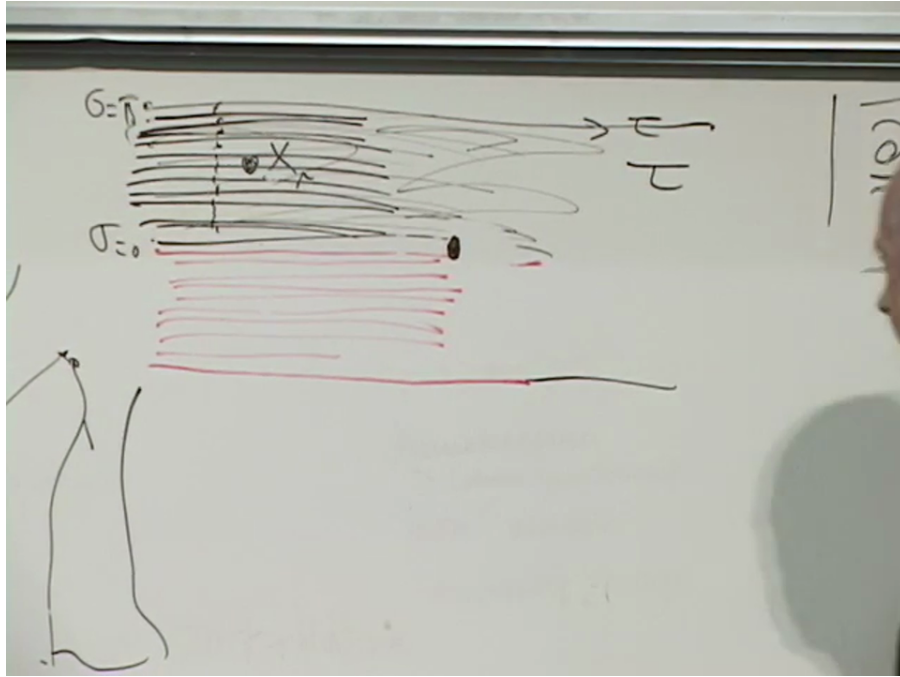


Figure 1.8: Two discretized open strings touch, form one intermediate string for a duration τ , and split. The source frame preserves the spring-and-worldline construction; the schematic isolates its connectivity.

Lecture frame: 00:51:25.

The two strings are drawn adjacent in parameter space only. Their spacetime positions need not be adjacent until the endpoint coordinates actually coincide. Reparameterization freedom lets the joining and splitting events be placed at convenient positions on the diagram; this geometric freedom will become conformal symmetry in the next lecture.

1.7 The Oscillator Calculation in Slow Motion

Let the first chain have coordinates $x_1, \dots, x_n$, and the second $x_{n+1}, \dots, x_{2n}$. Their center-of-mass coordinates are

$$X_{\text{cm}}^{(1)} = \frac{1}{n} \sum_{a=1}^n x_a, \quad X_{\text{cm}}^{(2)} = \frac{1}{n} \sum_{a=n+1}^{2n} x_a. \quad (1.21)$$

A chain with definite momentum k_1 has a center-of-mass plane wave multiplied by an internal oscillator ground state:

$$\Psi_1 = e^{ik_1 \cdot X_{\text{cm}}^{(1)}} \psi_0(x_1, \dots, x_n), \quad (1.22)$$

$$\Psi_2 = e^{ik_2 \cdot X_{\text{cm}}^{(2)}} \psi_0(x_{n+1}, \dots, x_{2n}). \quad (1.23)$$

The factors ψ_0 depend on relative coordinates, not on the common translation of an entire chain. Because the incoming strings are initially separate quantum systems,

$$\Psi_{\text{in}} = \Psi_1 \Psi_2. \quad (1.24)$$

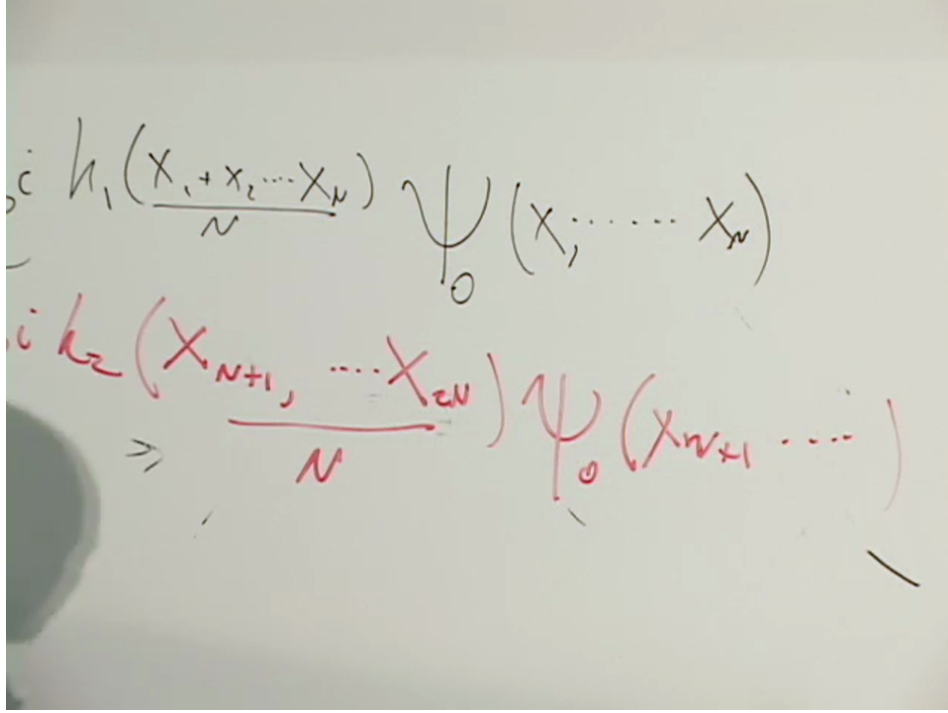


Figure 1.9: The two momentum plane waves and their internal oscillator ground-state factors, written in separate colors.

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Joining selects the part of this wavefunction for which the two endpoint coordinates agree:

$$x_n = x_{n+1}. \quad (1.25)$$

At that instant the other coordinates are unchanged. The selected state is then evolved with the Hamiltonian H_{joined} of the single longer chain:

$$\Psi(\tau) = e^{-iH_{\text{joined}}\tau} \Psi(0). \quad (1.26)$$

Finally one takes its overlap with a product of outgoing string states carrying momenta q_3, q_4 .

$$\psi_0(x_1, \dots, x_N) = e^{i h_1(\frac{x_1 + x_2 + \dots + x_N}{N})}$$

$$e^{i h_2(\frac{x_{N+1} + \dots + x_{2N}}{N})} \psi_0(x_{N+1}, \dots, x_{2N})$$

Figure 1.10: The two-chain wavefunction immediately before the endpoint coincidence is imposed.

Lecture frame: 00:58:55.

The calculation is lengthy but not mysterious. Every internal coordinate can be decomposed into oscillator modes, so the joining overlap, time evolution, and final projection reduce to Gaussian oscillator algebra. What remains unfixed is the lifetime τ of the joined string.

Question from the audience. Are the black and red wavefunctions added or multiplied? ^a

Response. Multiplied. The tensor-product state of two independent systems is represented in the coordinate basis by the product of their wavefunctions. Addition would describe a superposition of alternatives, not the simultaneous presence of both incoming strings.

^aLecture timestamp: 01:02:26.

Question from the audience. Why integrate over the intermediate time τ ? ^a

Response. The joined string is not assigned a fixed lifetime. Feynman's rule is to sum amplitudes over all allowed histories. In this description the remaining continuous history parameter is the time between joining and splitting, so the sum becomes an integral over $0 \leq \tau < \infty$.

^aLecture timestamp: 01:04:04.

1.8 From Propagation Time to the Beta Function

After evaluating the oscillator overlaps, the result can be organized as

$$A(s, t) \propto \int_0^\infty d\tau e^{\tau(s+1)} (1 - e^{-\tau})^{-t-1} e^{-\tau}. \quad (1.27)$$

The final $e^{-\tau}$ is kept visible because it is the Jacobian needed in the next step. Equivalently, the first and last exponentials combine to $e^{s\tau}$.

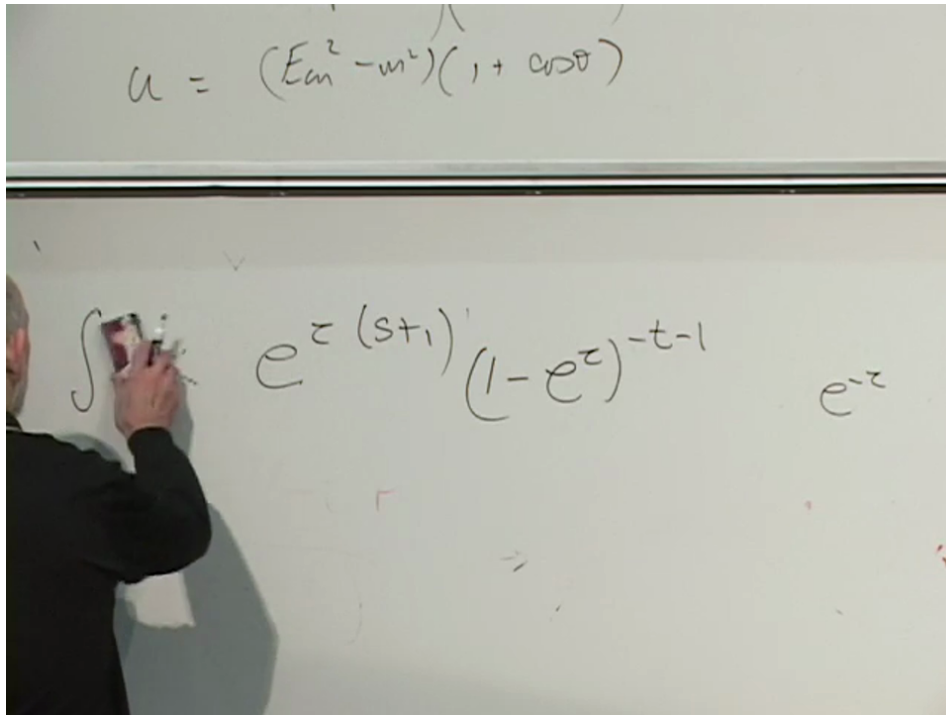


Figure 1.11: The propagation-time integrand after the oscillator calculation; the separate $e^{-\tau}$ factor supplies the change-of-variables Jacobian.

Lecture frame: 01:06:45.

Now set

$$z = e^{-\tau}, \quad dz = -e^{-\tau} d\tau. \quad (1.28)$$

As τ runs from 0 to ∞ , z runs from 1 to 0. The minus sign reverses the limits, giving

$$\mathcal{A}(s, t) \propto \int_0^1 z^{-s-1} (1-z)^{-t-1} dz. \quad (1.29)$$

The image shows a handwritten derivation on a chalkboard. It starts with the integral $\int_0^\infty e^{-\tau} (s-1)' (1-e^{-\tau})^{-t-1} d\tau$. Below this, the substitution $e^{-\tau} = z$ is written. The final integral is $\int z^{s+1} (1-z)^{-t-1} dz$.

Figure 1.12: The substitution $z = e^{-\tau}$ exposes the Euler beta integral.

Lecture frame: 01:08:49.

This integral is defined directly where its endpoints converge and elsewhere by analytic continuation. It is the Euler beta function:

$$B(-s, -t) = \frac{\Gamma(-s)\Gamma(-t)}{\Gamma(-s-t)}. \quad (1.30)$$

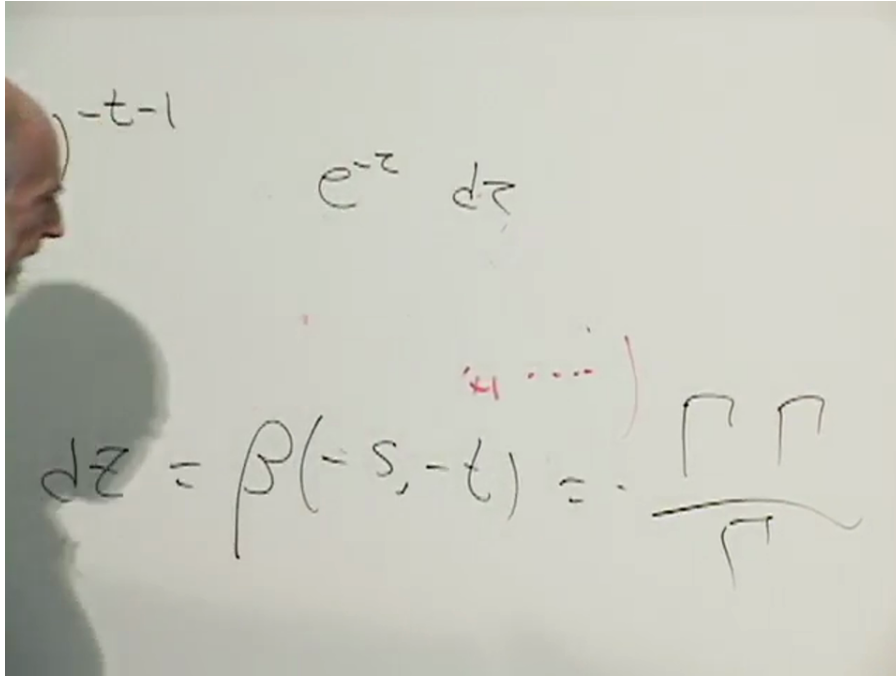


Figure 1.13: The board identifies the transformed integral with $B(-s, -t)$ and begins restoring its gamma-function form.

Lecture frame: 01:12:24.

The channel symmetry is now elementary. Under $z \mapsto 1 - z$,

$$\int_0^1 z^{-s-1} (1-z)^{-t-1} dz = \int_0^1 z^{-t-1} (1-z)^{-s-1} dz. \quad (1.31)$$

Thus $\mathcal{A}(s, t) = \mathcal{A}(t, s)$. A process constructed as two strings joining and later splitting automatically contains the crossed description as well. ²

Question from the audience. The gamma-function ratio resembles factorial expressions in combinatorics. Is there a direct physical counting interpretation? ^a

Response. At integer arguments the beta function is related to a ratio such as $n!m!/(n+m)!$, but no combinatorial mechanism is being asserted here. The same special function happens to arise from the worldsheet integral; this resemblance was not the route by which the amplitude was found.

^aLecture timestamp: 01:13:22.

The still-hidden reason why one worldsheet can be read in both channels is conformal symmetry. It permits the sheet to be stretched like taffy without changing the amplitude, turning a long direct-channel propagation region into a crossed-channel exchange picture. That geometric explanation is deferred to the next lecture.

1.9 What the Scattering Test Established

The discrete joining picture prompted a final cluster of questions.

²Editorial clarification: The lecture suppresses the Regge slope and intercept in the arguments of the beta function. In standard notation one replaces s, t by the corresponding linear trajectories $\alpha(s), \alpha(t)$. The duality and the change-of-variables argument are unchanged.

Question from the audience. If two chains of n constituents join at their endpoints, is the result $2n - 1$, and could that change fermion parity? ^a

Response. In the hadronic endpoint picture the touching endpoint pair may be regarded as a particle-antiparticle pair that annihilates, so the count is reduced by two, not one. In the continuum limit the finite difference between $2n$ and $2n - 2$ is immaterial, but preserving fermion parity is not. The lattice constituents are only a regulator for the continuous string, not literal fundamental beads.

^aLecture timestamp: 01:16:16.

Question from the audience. Will the later discussion continue into the full mathematics of superstrings? ^a

Response. Only selected consequences will be used. The immediate route is through worldsheet symmetry and compactification, followed by a different starting point developed in the mid-1990s: M-theory. Its purpose is to reach the same physics from a viewpoint in which several otherwise complicated features become transparent.

^aLecture timestamp: 01:18:05.

Question from the audience. Was this joining rule only a clue from meson collisions? ^a

Response. Meson scattering supplied the historical clue, but the rule is general for open strings. The same topology describes collisions of open-string photons and other open-string states. Two closed strings join and split by an analogous construction, with a closed string in the intermediate region.

^aLecture timestamp: 01:20:02.

Question from the audience. Does the calculation apply only when the string states are fermions? ^a

Response. No. The original Veneziano calculation belongs to the bosonic open string. Superstrings require additional fermionic degrees of freedom and modify details of the integrand, but the joining, propagation, projection, and worldsheet-duality logic remains closely analogous.

^aLecture timestamp: 01:20:56.

Finding a spin-one or spin-two state in the oscillator spectrum was suggestive, but not decisive. Their scattering amplitudes supplied the stronger test. Photon amplitudes obey the identities associated with charge conservation and gauge invariance; graviton amplitudes obey the corresponding constraints of a massless spin-two field. The string states passed these tests in interactions with massive and massless states and with one another. That is why the photon and graviton labels became physical identifications rather than guesses based only on polarization counting. ³

Question from the audience. Where does electric charge itself come from in the string description? ^a

Response. That question is left open at the end of the lecture. Its answer requires the internal and endpoint structure of the string theory, which has not yet been introduced, so it belongs to the discussion that follows rather than to this scattering derivation.

^aLecture timestamp: 01:24:03.

³Editorial clarification: In modern language these tests are encoded in decoupling of unphysical polarizations, gauge Ward identities, and the universal low-energy coupling of the massless spin-two state.